Modovate POC: Product Vision and Build Guide

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Section 1: What Modovate Is and Why This POC Matters

Homeowners in Ontario are sitting on $11 billion in available energy upgrade incentives but have no single platform to guide them from "I should probably upgrade my furnace" to "here is my plan, here are my options, here is what it costs after rebates, and here is a contractor I can trust." The market is fragmented: rebate programs are buried on government websites, contractor quality is inconsistent, and no one has built a consumer-grade guided decision tool for this space. The result is a massive pool of motivated homeowners making $10,000 to $50,000 decisions without the information they need, and a substantial stack of unclaimed public money sitting unused. Modovate fills that gap.

Modovate is a conversational AI platform, not an energy modeling engine. The product guides homeowners through a short intake, enriches their property data automatically, and delivers a personalized upgrade plan with real rebate amounts, curated equipment options, and matched contractors — all in one continuous experience. The homeowner enters their address and leaves with a complete project plan. That is the core product loop. The AI layer does not mean complex simulation or physics modeling. It means a smart, conversational interface that asks the right questions in plain language, synthesizes the answers with property data, and presents results in a format a non-engineer can act on.

Modovate is a two-sided marketplace. On one side: homeowners who want guidance. On the other: HVAC contractors, energy equipment suppliers, and eventually grant providers and financers. Revenue comes from contractor lead fees ($100 to $200 per qualified lead), product commissions (5 to 10% on equipment recommended and selected through the platform), and eventually white-label licensing to major home services retailers ($100K to $300K per year). The supply side — contractors and suppliers — is being built first, before homeowner demand is scaled. Ontario is the launch market because of the density of available government incentives and a well-documented regulatory environment. The business projects $3.95M in annual revenue by Year 3.

This POC is 90% an investor pitch demo and 10% internal alignment. An investor should be able to open the URL, spend 5 minutes clicking through the homeowner journey, and immediately understand what Modovate does and why it works as a business. Every screen

matters. The quality bar is high. "Done" means polished, fast, and fully navigable from address entry to the project summary dashboard, with no broken layouts, no Lorem ipsum, and no placeholder screens. The investor should finish the demo feeling like they just used a finished product, not a prototype.

Section 2: The Customer Journey We Are Building

We designed the full 8-screen customer journey in a wireframe flow. Here is a description of each screen in sequence, including what the homeowner sees and does, what makes each screen matter to an investor, and whether the data behind it is a real API call or mocked data.

Screen 1: Welcome / Address Entry

- **What the homeowner sees and does:** The visitor lands on a clean, full-screen landing page with the Modovate logo, a short positioning headline, and a single prominent address input field. As they begin typing a street address, Google Places Autocomplete surfaces matching Canadian addresses in real time. Once an address is selected, a satellite map preview of the property loads immediately below the input field. There is one call to action: "Get My Home Energy Report." The page asks for nothing else. No email, no navigation, no other choices.
- **What this screen communicates to an investor:** This establishes in under five seconds that Modovate is a technology product, not a contact form. The satellite image of the user's actual home loading on screen is a visceral moment: the platform already knows where they live and is beginning to assess it.
- **Data:** Google Places Autocomplete (real API) for address suggestions; Google Maps JavaScript API with satellite view (real API) for the map thumbnail. These two are non-negotiable. They deliver the highest-impact visual moment in the demo at essentially zero cost.
- **Wireframe notes:** Clean hero layout. Address input field is centered and prominent. Satellite map thumbnail appears below the input as the address is entered. Single amber CTA button. Minimal footer. The visual simplicity signals focus: there is only one thing to do on this screen.

Screen 2: Conversational Intake

- **What the homeowner sees and does:** After address entry, the homeowner is guided through a short intake questionnaire presented in a conversational chat-bubble style.

Questions appear one at a time, left-aligned as if from an AI assistant. Answer options appear as large, clickable chips. Selecting a chip highlights it and triggers the next question after a short delay. The intake collects: current heating system, home size, utility types, annual utility spend, and the homeowner's primary goals. A progress bar is visible throughout. The final confirmation message is "Great. We're building your home energy report."

- **What this screen communicates to an investor:** This demonstrates the Modovate AI persona: a knowledgeable but approachable guide that asks smart, efficient questions rather than presenting a 40-field engineering form. The conversational design signals that this product was built for homeowners, not engineers.
- **Data:** All user input only. No API calls. Answers are stored in session state and passed to every subsequent screen.
- **Wireframe notes:** Chat-bubble style layout with Modovate AI on the left. Previously answered questions are visible above, greyed out, showing progress. Answer options render as large selectable chips. Progress bar visible at the top of the screen throughout.

Screen 3: Property Assessment Results

- **What the homeowner sees and does:** This screen presents the homeowner's property data, auto-populated without any additional input from them. They see an embedded Google Maps satellite view of their property at the top, taking up approximately 40% of the screen height. Below: a grid of six property data cards covering estimated square footage, year built, roof area, climate zone, estimated annual energy cost, and a current EnerGuide-style energy rating. A baseline energy rating bar below the cards gives a plain-language assessment of their home's current efficiency.
- **What this screen communicates to an investor:** This is the automation moment. The homeowner entered only their address and answered a few questions. The platform synthesized those inputs with property data to produce a building profile. This answers the investor question: "What does the AI actually do?"
- **Data:** Google Maps JavaScript API satellite view (real, reused from Screen 1). All property data (square footage, year built, footprint, roof area) is mocked, keyed to the entered address via a lookup table of 3 to 5 demo addresses with a plausible default for all others.
- **Wireframe notes:** Full-width satellite map at the top of the screen, approximately 40% of screen height. Below: a grid of property data cards. Horizontal baseline energy rating bar below the cards. The satellite view of the actual property is the visual centrepiece of this screen.

Screen 4: Upgrade Recommendations

- **What the homeowner sees and does:** Based on the property profile and intake answers, the platform presents a prioritized list of recommended upgrades in a two-column card grid. Each card has an upgrade category icon, a short value-proposition headline, an estimated cost range, an estimated annual savings figure, and a rebate availability badge in amber. The top three cards carry a "Recommended" or "High Priority" tag. At the bottom of the screen, a contractor match teaser section previews what is coming next. Clicking "Explore This Upgrade" on any card advances to Screen 5 for that category.
- **What this screen communicates to an investor:** This makes the platform's intelligence layer visible. The recommendations are personalized, not generic. The rebate badges immediately signal the financial opportunity and the platform's awareness of available programs.
- **Data:** No external API calls. All logic is rule-based and deterministic: gas furnace homes get Heat Pump at top priority; pre-1995 homes get Insulation; goals including EV charging trigger an EV Charger recommendation. Cost and savings figures come from the mock data lookup tables.
- **Wireframe notes:** "Your Personalized Upgrade Plan" header. Two-column grid of upgrade cards. Each card has icon, upgrade name, value prop headline, cost range badge, green savings badge, and amber rebate badge. Contractor match teaser section at the bottom prompts the homeowner to continue.

Screen 5: Equipment Options (Heat Pump)

- **What the homeowner sees and does:** When the homeowner selects the Heat Pump upgrade, they arrive at an equipment browsing screen showing three product cards side by side. Each card displays a product image area, product name, brand, spec chips (COP, BTU, HSPF2, operating temperature), installed cost range, rebate eligibility badge, and a star rating. One card is highlighted as "Best Match." Each card has "Add to Project" and "Learn More" buttons.
- **What this screen communicates to an investor:** This demonstrates the marketplace and commerce layer of Modovate. The homeowner is not sent to a contractor to figure out equipment; the platform presents curated, spec-ready options. This is where Modovate earns product commissions (5 to 10%) in the revenue model. The investor sees both the homeowner value and the business model in one screen.
- **Data:** All product data is loaded from `data/equipment-catalog.json` (mocked). No live supplier API. The three heat pump products — Mitsubishi Zuba-Central H2i, Lennox XP25-H, and Bosch IDS Ultra Quiet — use real product names, real specifications, and realistic Ontario market pricing. For the POC, only the heat pump category needs a fully built equipment screen.

- **Wireframe notes:** "Cold Climate Heat Pumps for Your Home" header. Three product cards side by side. Each card includes product image area, product name, brand, spec chips, installed cost, rebate eligibility, and star rating. One card highlighted as "Best Match" with "Add to Project" and "Learn More" CTAs on each card.

Screen 6: Rebate and Incentive Summary

- **What the homeowner sees and does:** After reviewing equipment options, the homeowner arrives at a rebate summary screen. At the top: a large hero figure showing total available rebates (e.g., "$13,400 in rebates available for your project"). Below: individual rebate program cards, one per program, each showing program name, administering body, maximum amount, and key eligibility condition. A running total at the bottom. A clear amber CTA to add rebates to the project plan.
- **What this screen communicates to an investor:** Rebate intelligence is one of Modovate's most significant differentiators. Most homeowners do not know these programs exist, let alone how to stack them. This screen makes the financial case for taking action: the net cost of the project is substantially lower than the sticker price, and the platform has done the research.
- **Data:** No live API calls. All rebate data is hardcoded from current Ontario and federal programs as of Q1 2026. See Section 5 for the exact program list. The matching logic is rule-based: if the heat pump upgrade has been added, all heat-pump-eligible programs are shown. The dollar amounts and program names must be real. Do not invent program names or amounts.
- **Wireframe notes:** Hero bar at top with total rebates figure prominently displayed. Below: rebate program cards with program name, administering body, maximum amount, and eligibility note. Running total at bottom. The dollar amount in the hero bar is the hook — it must be real and credible.

Screen 7: Contractor Match

- **What the homeowner sees and does:** This screen presents three pre-qualified contractor profile cards. Each card shows contractor name, service area, specialties, star rating, years in business, number of projects completed, and availability. One card is highlighted as "Top Match." A map thumbnail shows the Toronto area with service area pins. Each card has a primary "Request Quote" CTA that opens a pre-filled contact modal.
- **What this screen communicates to an investor:** This makes the supply-side marketplace visible. Investors understand the contractor lead-gen model instantly: Modovate has qualified contractors, the homeowner is warm, and a click generates a

$100 to $200 lead fee. The profile cards signal trust: these are vetted partners, not random search results.

- **Data:** Fully mocked. All three contractor profiles are fictional but realistic GTA businesses: GreenHome HVAC Solutions, EcoRenovate Ontario, and Volta Energy Systems. No live directory, no scheduling integration. The "Request Quote" modal does not require a backend submission for the POC.
- **Wireframe notes:** "Matched Contractors Near You" header. Three contractor profile cards, each with name, service area, specialties, star rating, years in business, projects completed, and availability. One card highlighted as "Top Match." Map thumbnail showing Toronto with contractor service area pins. "Request Quote" CTA on each card.

Screen 8: ROI / Project Summary Dashboard

- **What the homeowner sees and does:** This is the final screen and the most important one. Three large hero metric cards across the top display Total Project Cost, Net Cost After Rebates, and Annual Savings. Below: a payback timeline bar visualizing the break-even point on a 15-year horizon, and a 10-year cumulative savings chart. On the right: a CO2 reduction badge with plain-language context (e.g., "Equivalent to removing one car from the road each year") and the full upgrade checklist. At the bottom: three CTA buttons — "Get Contractor Quotes," "Apply for Rebates," and "Download Your Report."
- **What this screen communicates to an investor:** This is the closing argument. It closes the loop from "I entered my address" to "I have a complete project plan with a payback period and net cost." The homeowner arrived confused and leaves with clarity. No competitor delivers financial intelligence, rebate access, and a path to execution in a single continuous flow.
- **Data:** No external API calls. All values are calculated client-side using the mock data and the homeowner's intake answers. Calculations use formula-based lookups: payback period is net cost divided by annual savings; CO2 reduction uses Natural Resources Canada emission factors for Ontario's electricity grid. Different intake answers produce different numbers. See the full calculation spec in the build plan.
- **Wireframe notes:** Three large hero metric cards across the top (Total Cost, After Rebates, Annual Savings). Left column: payback timeline bar and 10-year savings chart. Right column: CO2 reduction badge and upgrade checklist. Three CTA buttons at the bottom.

Section 3: What Is Real and What Is Mocked

Here is the full breakdown of every data source across all 8 screens. Use this as the reference whenever you are deciding whether to wire up a real API or write a mock.

Screen	Data Source	Real or Mocked	Rationale
1: Welcome	Google Places Autocomplete	Real API	Core UX, low cost, high visual impact
1: Welcome	Google Maps Satellite	Real API	Core differentiator visual
2: Intake	User input	N/A	Pure UX, no data needed
3: Property Assessment	Google Maps Satellite (reused)	Real API	Reuses Screen 1 tile
3: Property Assessment	Property data (sq ft, year built, etc.)	Mocked	MPAC integration not needed for POC
4: Recommendations	Rule-based logic	Mocked	Demonstrating personalization, not algorithm accuracy
5: Equipment Options	Product catalog	Mocked JSON	Supplier API agreements not in scope for POC
6: Rebate Summary	Rebate program data	Hardcoded real data	No rebate API exists; real values used for credibility
7: Contractor Match	Contractor profiles	Fully mocked	Contractor onboarding is a sales function, not a tech function
8: ROI Dashboard	Formula-based calculations	Mocked lookup table	EnergyPlus simulation not appropriate for POC

Why We Mock What We Mock

The two Google APIs are non-negotiable real integrations. Google Places Autocomplete (Screen 1) and the Google Maps JavaScript API satellite view (Screens 1 and 3) are real, live API calls. They deliver the single highest-impact visual moments in the demo — address autocomplete feels like a real technology product, and the satellite image of the homeowner's actual property is the "wow moment" that makes investors believe. Both APIs cost fractions of a cent per use and require minimal setup. There is no reason to mock them and every reason to implement them properly.

Everything else is mocked because: (a) real integrations add setup time and failure risk during live demos — a dependency on a live supplier API or government database is a liability, not an asset, during an investor walkthrough; (b) a well-crafted JSON file is indistinguishable from a live feed to an investor audience — they cannot tell whether the product data is coming from a database or a static file; (c) the goal of this POC is demonstrating the product concept, not proving production infrastructure.

Real program names, real dollar amounts, real URLs: this is non-negotiable for the rebate data. The rebate programs on Screen 6 must use actual current Ontario and federal program names, actual maximum amounts, and real application URLs. The programs are listed in detail in Section 5. Do not invent program names or make up dollar amounts. A knowledgeable investor — which is exactly who this demo is for — will recognize fake rebate data immediately, and it will undermine the credibility of the entire demo.

Section 4: Tech Stack Guidance

The following is our suggested approach for the tech stack. These are reasoned recommendations based on what will get us to a polished demo fastest. Where we say non-negotiable, we mean it — those decisions have direct investor demo impact. Everything else is your call as the developer.

Frontend Framework

Recommendation: Next.js with App Router.

Next.js gives us server-side rendering for fast initial page loads, built-in API routes so we do not need a separate backend server, and one-command deployment to Vercel. The App Router (Next.js 14+) is the current standard.

React with Vite is also acceptable if you prefer it. The difference is minimal for a POC. The one practical reason to lean toward Next.js is that SSR loads the Google Maps embed faster on first render, which matters for the demo experience on Screen 3.

Styling

Recommendation: Tailwind CSS + shadcn/ui.

Tailwind is strongly recommended for fast iteration and built-in responsive design. It keeps the codebase clean and avoids the stylesheet sprawl that happens with custom CSS at speed.

shadcn/ui is a good pairing: it provides pre-built, polished components (cards, buttons, modals, badges) that work directly with Tailwind, without requiring a heavy component library. This covers the UpgradeCard, RebateCard, ContractorCard, and modal patterns without writing everything from scratch.

Conversational Intake (Screen 2)

A custom step-wizard component is all that is needed. This is NOT an LLM-backed chat interface. The questions are hardcoded and deterministic. Each step renders a question as a styled chat bubble and the answer options as selectable chips. The logic advances based on the selected chip, stores the state in React context or sessionStorage, and moves to the next step after a short delay (approximately 400ms) to give the interaction a conversational feel.

Optional: a single LLM API call to OpenAI GPT-4o-mini or Gemini Flash to generate a personalized greeting using the entered address: "I found your property at [address]. Let me ask you a few questions." This adds a nice touch but is not required for the POC to be investor-ready. Build the deterministic wizard first; add the LLM greeting only if time permits.

Backend

No separate backend server is needed. All server-side logic — geocoding, property data lookup, calculation logic — lives in Next.js API routes, which are serverless functions. These run on Vercel automatically at no extra cost and require no configuration. Do not introduce a FastAPI, Express, or Node server process alongside the Next.js app.

Non-Negotiable APIs

There are exactly two real API integrations required, and both are non-negotiable:

1. **Google Places API:** Powers the address autocomplete on Screen 1. This is not optional. The address input must feel like a real search — Canadian address suggestions appearing as you type — not a plain text field. A plain input field would undermine the entire "technology product" impression on the first screen.

2. **Google Maps JavaScript API (satellite view):** Powers the satellite map on Screens 1 and 3. This is not optional. The satellite image of the homeowner's actual property is the single highest-impact visual moment in the demo. It makes the product feel real in a way that no other element does. This must work for any Ontario address, not just a hardcoded set.

Full setup steps for both are in Section 7.

Mock Data

All product, rebate, and contractor data lives in static JSON files in the `/data` directory. No database is needed. No ORM. No PostgreSQL. The application reads these files at build time or request time via Next.js API routes. Energy load and savings calculations are TypeScript functions using simple lookup-table formulas — no simulation engine.

Session State

Use React context (a `HomeownerContext` provider wrapping the app in `layout.tsx`) synced to `sessionStorage` on every state update. Answers from Screen 2 are passed through to Screens 3 through 8 via this context. No persistence between sessions is required. No database.

On initial load, rehydrate from `sessionStorage` if data exists — this allows page refreshes during development without losing state, which saves significant time.

Hosting

Vercel. Non-negotiable for demo purposes.

One command to deploy, instant shareable URL, the free tier is sufficient for a demo application. The URL should be clean: `modovate-demo.vercel.app` or a custom domain if available. The investor receives a URL they can open on any device. There is no local setup, no Docker, no SSH.

What to Avoid at This Stage

- **Production-grade simulation engines** (EnergyPlus, OpenStudio): not needed; adds weeks of setup for zero investor-visible value
- **Authentication or user accounts**: the demo is entirely anonymous
- **Separate backend servers** (FastAPI, Express, Node): Next.js API routes handle everything
- **Complex databases** (PostgreSQL, TimescaleDB): JSON files are sufficient
- **Docker or containerization**: Vercel handles deployment; no containers needed

- **LangGraph, LangChain, or any multi-agent orchestration**: the intake is a deterministic step wizard, not an agentic pipeline

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Section 5: Ontario Rebate Programs (Use These Exactly)

These are the real, current Ontario and federal rebate programs as of early 2026. Use these exact program names, dollar amounts, and URLs in `mock-rebate-programs.json` and on Screen 6. Do not invent program names or amounts.

Federal Programs

Canada Greener Homes Grant

- Administering body: Natural Resources Canada (NRCan)
- Maximum amount: $5,000 total per household
- Eligible upgrades: Air-source heat pumps (up to $5,000), insulation (up to $600 per zone, up to $5,000 total), windows and doors, solar PV
- Key condition: Requires a pre- and post-retrofit EnerGuide assessment by an NRCan-registered energy advisor. For the POC demo, display the rebate amount and note the assessment requirement; do not simulate the assessment process.
- Application URL: https://www.nrcan.gc.ca/energy-efficiency/homes/canada-greener-homes-grant/23230

Canada Greener Homes Loan

- Administering body: NRCan
- Maximum amount: $40,000 interest-free loan, up to 10-year term
- Eligible upgrades: Same as the Grant
- Key condition: Must complete a pre-retrofit EnerGuide assessment
- Note: This is a loan, not a grant. Display separately on Screen 6 to avoid confusion with the grant amount.
- Application URL: https://www.nrcan.gc.ca/energy-efficiency/homes/canada-greener-homes-loan/24286

Ontario / Utility Programs

Enbridge Home Efficiency Rebate Plus (HER+)

- Administering body: Enbridge Gas (Ontario)
- Maximum amount: Up to $10,000 for eligible retrofits
- Eligible upgrades: Air-source heat pumps (up to $6,500), insulation, air sealing, smart thermostats, heat pump water heaters
- Key condition: Must be an Enbridge Gas customer; pre-retrofit home assessment required
- Stackable with: Canada Greener Homes Grant (programs are designed to be stacked)
- Application URL: https://www.enbridgegas.com/home-efficiency-rebate

IESO saveONenergy Programs

- Administering body: IESO / local electric utility
- Programs include: Home Assistance Program (low-income focus), Retrofit Program, Peak Saver Plus
- Amounts: Typical residential rebates range from $50 to $500 for appliances and thermostats; larger amounts for eligible insulation and HVAC work
- Key condition: Must be a customer of a participating Ontario electric utility
- Application URL: https://www.ieso.ca/consumer-partnership/energy-efficiency-programs

Toronto Hydro Energy Conservation Programs

- Administering body: Toronto Hydro (Toronto customers only)
- Programs: peaksaver PLUS (smart thermostat rebate), Home Energy Savings Program
- Amounts: Smart thermostat rebate up to $75; additional conservation program rebates vary
- Key condition: Toronto Hydro customer
- Display rule: Show this program only when the entered address is in the City of Toronto. For all other Ontario addresses, display the generic IESO programs instead.
- Application URL: https://www.torontohydro.com/for-home/energy-conservation

Ontario Renovates Program

- Administering body: Ontario Ministry of Municipal Affairs and Housing
- Maximum amount: Forgivable loans up to $25,000 for eligible homeowners
- Key condition: Income-qualified; priority for Indigenous, Northern, and rural homeowners
- Display note: Include this program in the summary with an eligibility note. Do not assume eligibility for the demo profile; display it as "Check eligibility" rather than "You qualify."
- Application URL: https://www.ontario.ca/page/get-help-with-home-repairs-and-modifications

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Stacking Note for the Demo

For a typical GTA homeowner replacing a gas furnace with a cold-climate heat pump and adding insulation:

- Canada Greener Homes Grant: up to $5,000
- HER+ (Enbridge): up to $6,500 for heat pump plus up to $2,000 for insulation
- IESO/utility programs: up to $500
- **Total stacked rebates: up to $14,000**

This stacked total is what the "Total available rebates" figure on Screens 6 and 8 should reflect for a heat pump plus insulation project on a typical demo profile.

Section 6: Mock Data Specifications

The following files need to be created before the first demo can run. Each file has a defined structure and a target record count. Commit all files to the `/data` directory in the repository.

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1. Equipment Catalog

File: `data/equipment-catalog.json`

A JSON object with categories as keys, each containing a label, description, and products array. Required for Screen 5.

Required fields per product:

```
{

  "id": "string",

  "name": "string",

  "brand": "string",

  "specs": {

    "cop": "number",
```

```
    "btu": "number",

    "hspf2": "number",

    "minOperatingTemp": "number (Celsius)"

  },

  "installedCostMin": "integer (CAD)",

  "installedCostMax": "integer (CAD)",

  "rebateEligible": "boolean",

  "rebatePrograms": ["array of program ID strings"],

  "bestMatch": "boolean",

  "highlights": ["array of 2-3 plain-language feature strings"]

}
```

Target quantity: 3 to 5 products per category, across 7 categories. Approximately 25 to 30 total product records.

Categories and products to include:

- Heat Pump: Mitsubishi Zuba-Central H2i, Lennox XP25-H, Bosch IDS Ultra Quiet
- Insulation: Closed-cell spray foam (Demilec), Blown-in cellulose, Rigid foam board (Rockwool ComfortBoard)
- Electrical Panel: Siemens 200A panel, Square D QO 200A panel, Eaton BR 200A panel
- Solar PV: LG NeON R 400W, Canadian Solar HiKu6 420W, SunPower Maxeon 6
- EV Charger: Flo G5 (32A), ChargePoint Home Flex, Grizzl-E Classic
- Battery Storage: Tesla Powerwall 3, LG RESU16H Prime, Sonnen Eco 10
- Windows: Pella 350 Series triple-pane, JELD-WEN Premium Vinyl, Marvin Essential double-hung

Heat pump product reference values (use these exactly for Screen 5):

```
{

  "id": "mitsubishi-zuba",
```

"name": "Mitsubishi Zuba-Central H2i",
"brand": "Mitsubishi Electric",
"specs": { "cop": 3.5, "btu": 36000, "hspf2": 10.5, "minOperatingTemp": -30 },
"installedCostMin": 18500,
"installedCostMax": 21000,
"rebateEligible": true,
"rebatePrograms": ["greener-homes"],
"bestMatch": true,
"highlights": ["Rated to -30C", "Qualifies for Greener Homes Grant", "Most popular in GTA"]
}
{
"id": "lennox-xp25h",
"name": "Lennox XP25-H",
"brand": "Lennox",
"specs": { "cop": 3.2, "btu": 42000, "hspf2": 9.8, "minOperatingTemp": -25 },
"installedCostMin": 17200,
"installedCostMax": 19500,
"rebateEligible": true,
"rebatePrograms": ["greener-homes"],
"bestMatch": false,
"highlights": ["Higher BTU capacity", "Lower upfront cost", "Wide dealer network"]

}

{

"id": "bosch-ids-ultra",

"name": "Bosch IDS Ultra Quiet",

"brand": "Bosch",

"specs": { "cop": 3.6, "btu": 30000, "hspf2": 11.0, "minOperatingTemp": -30 },

"installedCostMin": 20500,

"installedCostMax": 23000,

"rebateEligible": true,

"rebatePrograms": ["greener-homes"],

"bestMatch": false,

"highlights": ["Highest efficiency rating", "Ultra-quiet operation", "10-year parts warranty"]

}

---

2. Contractor Profiles

File: `data/contractor-profiles.json`

A JSON object with a `contractors` array. Required for Screen 7.

Required fields per profile:

{

"id": "string",

"name": "string",

"city": "string",

```
  "serviceArea": "string",

  "serviceAreaCities": ["array of city strings"],

  "specialties": ["array of category ID strings"],

  "rating": "number (one decimal, e.g. 4.9)",

  "reviewCount": "integer",

  "yearsInBusiness": "integer",

  "projectsCompleted": "integer",

  "availability": "string (e.g. 'Available within 2 weeks')",

  "certifications": ["array of certification strings"],

  "topMatch": "boolean",

  "description": "string (2-3 sentences, plain language)"

}
```

The three profiles to use (use these exactly):

1. **GreenHome HVAC Solutions** — Toronto, serves GTA broadly. Specialties: heat pump, electrical, insulation, air sealing. Rating: 4.9 (312 reviews), 14 years in business, 1,240 projects. Available within 2 weeks. Top Match. Certifications: HRAI Certified, Enbridge Registered Contractor, NRCan Registered.

2. **EcoRenovate Ontario** — Toronto, serves GTA + Hamilton. Specialties: heat pump, insulation, solar PV, windows. Rating: 4.8 (187 reviews), 8 years in business, 620 projects. Available within 3 weeks. Certifications: HRAI Certified, NRCan Registered, SunPower Elite Dealer.

3. **Volta Energy Systems** — Toronto East, serves Toronto East + Scarborough. Specialties: solar PV, EV charging, battery storage, electrical. Rating: 4.7 (94 reviews), 5 years in business, 280 projects. Available within 1 week. Certifications: ESA Licensed Electrical Contractor, Tesla Powerwall Certified, SolarEdge Installer.

These are fictional companies. Do not use the name of any real contractor.

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3. Rebate Programs

File: `data/rebate-programs.json`

A JSON object with a `programs` array. Required for Screens 6 and 8. Program values must match the programs listed in Section 5 exactly.

Required fields per program:

```
{

  "id": "string",

  "name": "string",

  "administrator": "string",

  "level": "string (federal | provincial | utility)",

  "maxAmount": "integer (CAD)",

  "eligibleUpgrades": ["array of category ID strings"],

  "perUpgradeLimits": {

    "category_id": "integer (max CAD for that category)"

  },

  "applicationUrl": "string (real URL)",

  "notes": "string (1-2 sentences, plain language eligibility note)",

  "active": "boolean",

  "province": "string (all | ON | etc.)",

  "utilityRestriction": "string or null (e.g. enbridge_gas_customer | toronto_hydro_customer | null)",
```

```
  "isLoan": "boolean (true only for the Greener Homes Loan and Toronto Hydro loan)"

}
```

Target quantity: 5 to 7 records covering the programs in Section 5. Refer to Section 5 for all program names, amounts, and URLs.

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4. Energy Savings Lookup Table

File: `data/mock-energy-savings.json`

A JSON lookup table used to calculate estimated annual savings and CO2 reduction on Screen 8. This maps a combination of home profile characteristics and upgrade type to a savings estimate, without requiring any simulation engine.

Dimensions per record:

```
{

  "upgradeCategory": "string (e.g. heat_pump, insulation)",

  "heatingSystemReplaced": "string (e.g. gas_furnace | oil_furnace | electric_baseboard)",

  "homeSizeBucket": "string (small | medium | large)",

  "homeAgeBucket": "string (pre_1980 | 1980_to_2000 | post_2000)",

  "annualSavingsLowCad": "integer",

  "annualSavingsHighCad": "integer",

  "co2ReductionTonnesPerYear": "number (one decimal)"

}
```

Size buckets: small = under 1,200 sq ft; medium = 1,200 to 2,000 sq ft; large = over 2,000 sq ft.

Target quantity: Approximately 20 to 40 records covering the most common combinations.

Reference sources for realistic values: Natural Resources Canada Home Energy Saver tool; Pembina Institute Ontario retrofit savings data. For a gas furnace to heat pump conversion in a

medium Ontario home built 1980 to 2000, a savings range of $900 to $1,400/year and approximately 3.2 tonnes CO2/year reduction is realistic.

CO2 calculation formulas used in `lib/energy-calculator.ts`:

- Electricity savings component: electric savings in kWh multiplied by 0.029 kg CO2/kWh (Ontario grid emission factor from NRCan)
- Gas savings component: estimated gas savings of 1,500 m3/year for a gas-to-heat-pump conversion, multiplied by 2.2 kg CO2/m3
- Sum both components, display in tonnes (divide total kg by 1,000)
- If no gas heating detected from intake answers, use electricity savings only

---

5. Demo Address Property Profiles

File: `data/mock-address-profiles.json`

A JSON object mapping specific demo addresses to pre-built property profiles. Required for Screen 3. This file ensures that when a specific address is entered during an investor demo, the property data displayed is credible and matches the actual satellite image the investor sees.

Required fields per profile:

```
{

  "address": "string (full standardized address)",

  "estimatedSqft": "integer",

  "yearBuilt": "integer",

  "footprintSqm": "integer",

  "roofAreaSqm": "integer",

  "climateZone": "string (Zone 5 | Zone 6)",

  "hasGarage": "boolean",

  "heatingSystemAssumption": "string (used as a pre-fill suggestion, not forced on the user)"

}
```

Target quantity: 3 to 5 demo addresses in the Toronto / GTA area. Select addresses covering a range of building profiles, for example:

- A 1960s bungalow in Scarborough
- A 1990s detached in Mississauga
- A 2005 semi-detached in North York

Select real street addresses in ordinary residential neighborhoods. Do not use the personal residences of any team members.

Default fallback profile (for any address not in the lookup table):

```
{

  "estimatedSqft": 1400,

  "yearBuilt": 1987,

  "footprintSqm": 125,

  "roofAreaSqm": 110,

  "climateZone": "Zone 5",

  "hasGarage": true,

  "heatingSystemAssumption": "gas_furnace"

}
```

This fallback ensures that the demo works with any Ontario address an investor might type, not just the pre-loaded set.

Section 7: Google Maps API Setup

Follow these steps exactly. Estimated time: 30 minutes.

1. Go to console.cloud.google.com and create a new project named "Modovate POC." Note the project ID.

2. In the APIs and Services library, enable all four of the following: Maps JavaScript API, Places API (use the legacy Places API, not Places API (New), to avoid SDK breaking changes), Geocoding API, and Maps Embed API.

3. Go to "Credentials" and click "Create Credentials," then "API Key." Copy the key.

4. Click "Restrict Key" and apply the following restrictions:

   - Application restrictions: set to "HTTP referrers (websites)"
   - Add `localhost:3000/*` for local development
   - Add your Vercel deployment URL (e.g., `modovate-poc.vercel.app/*`) after the first deploy
   - API restrictions: restrict to the four APIs listed in step 2 only

5. Create `.env.local` in the project root (this file must never be committed; add it to `.gitignore`):

   GOOGLE_MAPS_API_KEY=your_key_here

   NEXT_PUBLIC_GOOGLE_MAPS_API_KEY=your_key_here

   Both variables hold the same key. `GOOGLE_MAPS_API_KEY` is used in server-side API routes. `NEXT_PUBLIC_GOOGLE_MAPS_API_KEY` is accessible in client-side components (the `NEXT_PUBLIC_` prefix is required for Next.js to expose environment variables to the browser).

6. Install the Google Maps JS loader package:

   npm install @googlemaps/js-api-loader

7. Use the loader in `AddressInput.tsx` to initialize the Places Autocomplete library:

```
import { Loader } from '@googlemaps/js-api-loader';

const loader = new Loader({

  apiKey: process.env.NEXT_PUBLIC_GOOGLE_MAPS_API_KEY!,

  version: 'weekly',

  libraries: ['places'],

});
```

Call `loader.load()` inside a `useEffect` on component mount. On `place_changed`, extract `formatted_address`, `geometry.location.lat()`, and `geometry.location.lng()`, then store all three in `HomeownerContext`.

Restrict autocomplete to Canadian addresses: `componentRestrictions: { country: 'ca' }` and `types: ['address']`.

8. For the satellite map embed on Screens 1 and 3, use a plain `<iframe>` with the Google Maps Embed API — no JavaScript SDK needed for the embed view:

   ```
   const mapSrc =
   `https://www.google.com/maps/embed/v1/view?key=${process.env.NEXT_PUBLIC_GOOGLE_MAPS_API_KEY}&center=${lat},${lng}&zoom=19&maptype=satellite`;

   return <iframe src={mapSrc} width="100%" height="400" style={{ border: 0 }} allowFullScreen />;
   ```

   On Screen 1, use a smaller static thumbnail. On Screen 3, display the full interactive embed.

9. Add `.env.local` to `.gitignore`. The API key must never appear in the repository.

10. After the first Vercel deploy, go to your Vercel project settings and add `NEXT_PUBLIC_GOOGLE_MAPS_API_KEY` as an environment variable with the same key value. Add the Vercel domain to the API key's HTTP referrer restrictions in Google Cloud Console. Redeploy for the environment variable to take effect.

11. **Cost note:** Enable billing on the Google Cloud project (required to use the APIs, even at free tier). The Maps JavaScript API, Places API, Geocoding API, and Maps Embed API all have generous free tiers. Google provides $200/month in free credits. For a demo application with fewer than 1,000 sessions per month, the total cost will be $0.

Section 8: Definition of Done

The POC is investor-ready when every item on this checklist is true. Use this list to self-check before the first investor walkthrough.

Functional

- All 8 screens are navigable in sequence (click through the full homeowner journey without any broken links or missing routes)
- Address autocomplete works using the Google Places API (real Canadian address suggestions appear as you type)
- Selecting a Toronto or GTA address loads a satellite map of that property on Screen 1 and Screen 3
- Intake answers (Screen 2) feed into the recommendation engine: a gas furnace home and an electric baseboard home produce different recommendations on Screen 4
- Screen 5 shows at least 3 equipment options for the Heat Pump category with realistic specs, pricing, and rebate badges
- Screen 6 shows at least 3 rebate programs with real program names, real dollar amounts, and a running total that reflects the selected upgrades (not a hardcoded number)
- Screen 7 shows 3 contractor profiles with realistic GTA details and a "Request Quote" CTA that opens and closes a modal
- Screen 8 shows total cost, net cost after rebates, annual savings, and payback period all derived from the homeowner's inputs — different intake answers produce different numbers

Visual Quality

- All 8 screens use a consistent color palette, typography, and spacing (no visual drift between screens)
- No placeholder text anywhere (no "Lorem ipsum", no "[INSERT PRODUCT NAME]", no "TODO", no "[TBD]")
- No broken images or missing icons
- The design looks polished and professional at first impression, not like a prototype
- Works correctly on desktop Chrome and Safari (primary investor demo environment)
- Works correctly on a mobile browser at 390px viewport width (iPhone 15 Pro width): layout stacks correctly, no horizontal scrolling, no clipped content
- Page transitions load in under 2 to 3 seconds per screen advance on a standard broadband connection

Demo-ability

- Deployed on Vercel with a shareable URL (no local setup required by the investor)
- A first-time visitor with no instructions can navigate the full journey without help from the Modovate team
- The full journey (Screen 1 to Screen 8) takes 3 to 5 minutes to complete at a normal pace

- The "Request Quote" CTA on Screen 7 responds (opens a modal — a form that does not need to submit to a backend)
- The "Download Report" CTA on Screen 8 responds (triggers browser print, a static PDF, or a placeholder page — something must happen)

Not Required for Done

- Real contractor directory or live contractor data
- Live equipment supplier API
- Live rebate eligibility API (NRCan, Enbridge, IESO)
- User accounts or email capture
- Session persistence across devices or browsers
- A/B testing
- Analytics beyond basic Vercel Analytics (which is optional and takes 5 minutes to add)

Section 9: What Not to Build

To keep scope clean and avoid the patterns from the previous SRS, here is an explicit list of things that should not appear in the codebase at this stage. If any of these appear in a pull request, stop and check with Mark before proceeding.

Simulation and Modeling

- EnergyPlus, OpenStudio, or any physics-based building simulation engine
- EPW or CWEC weather datasets
- Building geometry or IDF file generation
- Any simulation worker or compute job queue

Orchestration and AI

- LangGraph, LangChain, or any multi-agent orchestration framework
- Multiple AI agents or agent pipelines
- Stateful LLM checkpointing
- Streaming token rendering (if the optional personalized greeting is added to Screen 2, basic SSE is acceptable, but no agent pipeline is needed)

Infrastructure

- Redis or any task/message queue (Celery, Bull, BullMQ, or equivalent)
- Docker or Docker Compose
- Ubuntu server setup, SSH-based deployment, or Nginx
- Any separate backend server process (FastAPI, Flask, Express, or standalone Node)

Database

- PostgreSQL, MySQL, or any relational database
- TimescaleDB, InfluxDB, or any time-series database
- pgvector or any vector database
- Any ORM (Prisma, Drizzle, SQLAlchemy, or equivalent)

Auth and Accounts

- JWT authentication
- OAuth flows or any third-party identity provider
- User registration or login screens
- Session persistence across devices

Out-of-Scope Features

- Insurance integration of any kind
- Financing or loan product integration (displaying the Greener Homes Loan as an informational card on Screen 6 is fine; building any loan application flow is not)
- New construction logic (Modovate is for existing homes only)
- Commercial or multi-unit building logic
- Automated government rebate application submission
- A/B testing infrastructure
- Multi-language support

Closing

Arjyabir, this document supersedes the SRS direction you received from the previous specification (Modovate_SRS-1-1.pdf, Elefusion Inc., March 2026). That document was built around an engineering-first framing that does not match what we need for this stage. This guide is the authoritative reference for everything going forward. All decisions about scope, architecture, and data should be made against this document.

The wireframe flow for all 8 screens is being shared alongside this guide. Use it as the visual reference for each screen's layout — it shows the specific arrangement of components, card grids, hero elements, and CTAs that we agreed on. The wireframes and this guide together should give you everything you need to start without any ambiguity.

I am available before development starts to walk through any part of this. If anything is unclear, if you see a technical reason to deviate from a recommendation, or if something in the scope seems underspecified, please flag it before you start building. I would much rather spend 30

minutes on a call resolving questions upfront than have us go in different directions for a week. Reach me directly at any time.

Looking forward to seeing this come together.

Mark Co-founder, Modovate April 2026